Conceptual design of Aircraft
Low speed Aerodynamics - Recap

Objectives:

1. Recap the definitions of lift and drag coefficients for an aircraft and an aerofoil.
2. Recap the concepts of linear aerodynamics and stall
3. Recap the concepts of base and induced drag

Conceptual design of Aircraft
Low speed Aerodynamics - Recap

- **Parameters:**
  - **Lift and drag coefficients:**
    - **3D body, i.e. wing:**

$$C_L \equiv \frac{L}{0.5\rho V^2 S_w}, \quad C_D \equiv \frac{D}{0.5\rho V^2 S_w}$$

    - **2D body, i.e. aerofoil:**

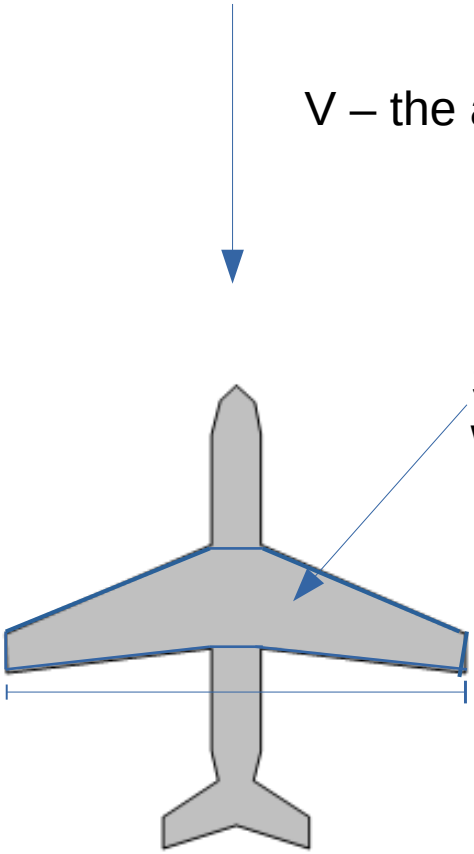
$$C_l \equiv \frac{L'}{0.5\rho V^2 c}, \quad C_d \equiv \frac{D'}{0.5\rho V^2 c}$$

  - **Wing load:**

$$w \equiv \frac{W}{S_w}$$ **with units of N/m^2**

S_W is the surface area of the wings (also noted simply as S)
c is the chord length of the aerofoil, i.e the distance from the leading edge to the trailing edge.

Let’s look in more details



We will define the lift coefficient C_L and drag coefficient C_D as:

$$C_L = \frac{L}{0.5\,\rho V^2 S}$$

$$C_D = \frac{D}{0.5\,\rho V^2 S}$$

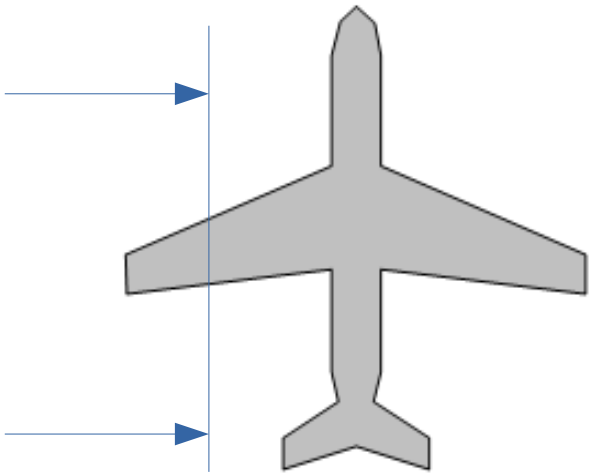
We will define the wings’ aspect ratio AR as:

$$AR = \frac{b^2}{S}$$

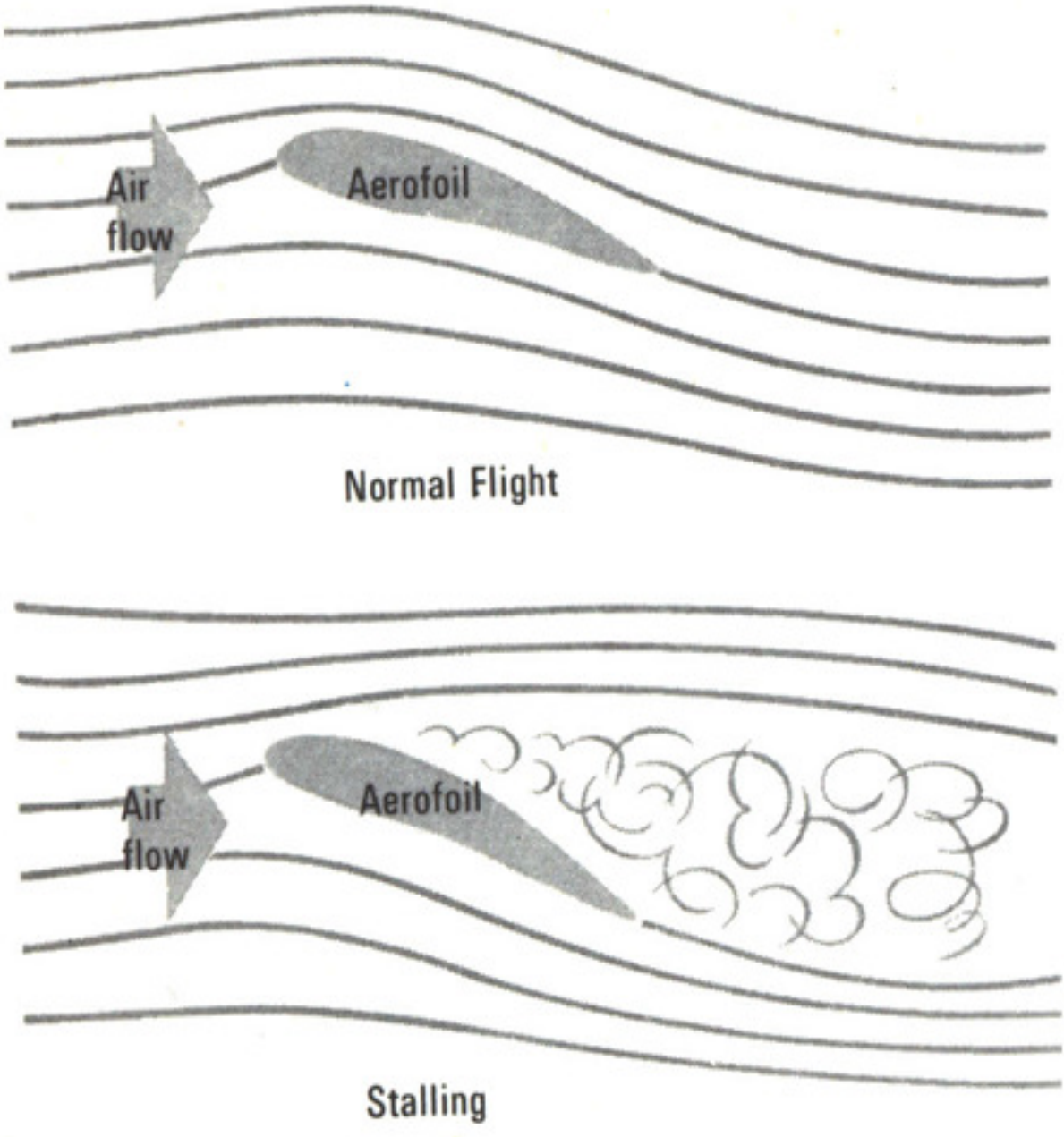
C_L, C_D and AR have no units!

Aerofoil concept

We will look at the cross-section of the wing. This cross-section is called aerofoil



Flow over an Aerofoil

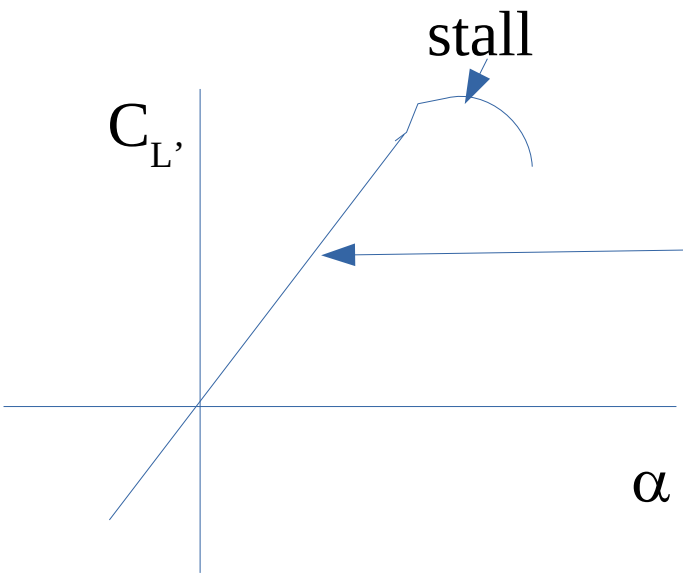


Lift is generated by pushing the flow down due to the geometry of the aerofoil. So by Newton's third law the aerofoil experiences opposite force, which is lift.
However, when the flow massively separates in stall and the flow does not follow the aerofoil's geometry, lift is lost and drag significantly increases (due to pressure drag)

Aerofoil's lift coefficient $C_{L'}$ and drag coefficient $C_{D'}$:

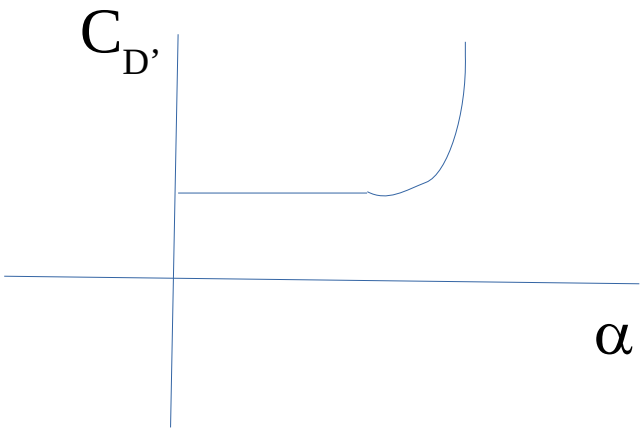
$$C_{L'} = \frac{L'}{0.5\,\rho V^2 c}\ ,\quad C_{D'} = \frac{D'}{0.5\,\rho V^2 c}$$

Units of L' and D' are (N/m)



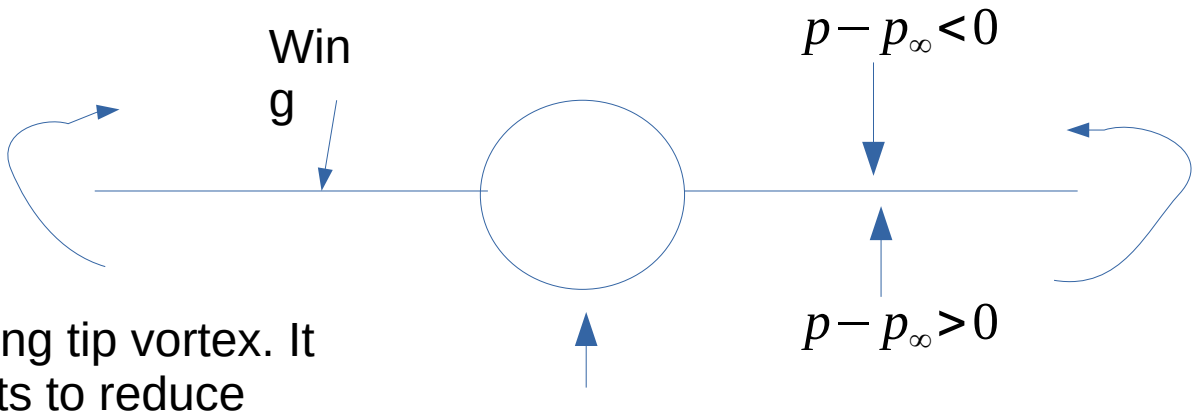
Symmetric profile

Sometimes the lift coefficient curve slope *a* is noted as $dC_{L'}/d\alpha$



Typically *a* = 5.7 (1/rad)

Finite wings effect



$D = D_0 + D_i$

D_0 – base drag (skin friction + form drag)

D_i – Induced drag (due to wings tips vortices)

We can also write $C_D = C_{D0} + C_{Di}$

C_{D0} – base drag coefficient

C_{Di} – induced drag coefficient

$$C_{Di} = \frac{k C_L^2}{\pi AR}, \quad k \geq 1$$

k = 1 for elliptic wings (as of the Spitfire)

Wing fences to reduce wing tip vortices and thus reduce induced drag



Canard configuration

Winglets to suppress wing tip vortices (and also to serve as vertical stabilisers)



The Raytheon Company's Beech Starship in a cruise condition

No induced drag on this electric bi-plane wings



The Modern Airliner

The Airbus Family





A 318

A 319

A 320

A 321

The Airbus 318



